

[Answer any five questions. Assume reasonable value for any missing data.]

Time: 3.00 Hours

- a) Define Orthogonal Matrix and Inverse Matrix. Show that every square matrix can be expressed as sum of the symmetric and Skew-symmetric matrix. 5
- b) If $AB = BA$ then prove that $(AB)^n = A^n B^n$ 6

c) Prove that $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary. 3

- a) Define matrix polynomial. Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ and $f(\lambda) = \lambda^2 - 4\lambda + 3$ then prove that A is a root of $f(\lambda)$. 3
- b) State and prove Cayley-Hamilton theorem. 5

c) Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$. 6

- a) What is curl of a vector? Prove that $\nabla \cdot (\nabla \times \mathbf{A}) = 0$. 4
- b) Find the equation of the tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(2, -1, 5)$. 6
- c) Find the directional derivative of $\phi(x, y, z) = 4xz^2 - 3x^2y^2z$ at the point $(2, -1, 2)$ in the direction $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. 4

a) Define Fourier series. If $f(x)$ is an even function then show that (i) $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$ 4

$$(ii) a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

b) Find the Fourier series expansion of the periodic function $f(x) = -x$ when $-\pi < x < 0$ and $f(x) = x$ when $0 < x < \pi$. Also prove that $\frac{\pi}{8} = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \dots$ 10

a) Define vector space and subspace. Show that $S = \{(a, b, c, d) \in \mathbb{R}^4 : 2a - 3b + 5c - d = 0\}$ is a subspace of $\mathbb{R}^4$. 5

b) Define linear dependence and independence of vector. Show that $\{(1, 1, -1), (1, 0, 2), (1, 1, 1)\}$ vectors are linearly independence. 6

c) Show that a single vector is linearly dependent and only if it is a zero vector. 3

a) Solve the initial value problem by Laplace transform $Y''' - 3Y'' + 3Y' - Y = t^2 e^t$; $Y(0) = 1, Y'(0) = 0, Y''(0) = -2$ 5

b) Using convolution theorem evaluate $L^{-1} \left\{ \frac{1}{s^2(s+1)^2} \right\}$ 5

c) If $L\{F(t)\} = f(s)$ then show that, $L\left\{ \int_0^t F(t-\tau) f(\tau) d\tau \right\} = \frac{f(s)}{s}$ 4

a) Show that $\mathbf{E} = (x^2 - yz)\mathbf{i} + (y^2 - zx)\mathbf{j} + (z^2 - xy)\mathbf{k}$ is a conservative force field. 2

b) Show that $\iint_S \mathbf{A} \cdot \mathbf{n} \, ds = 24$ where $\mathbf{A} = 18xz\mathbf{i} - 12z\mathbf{j} + 3y\mathbf{k}$ and S is the part of the plane $2x + 3y + 6z = 12$ which is located in the first octant. 5

c) Find the Fourier transform of $F(x) = \begin{cases} 1-x^2 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$ Also prove that $\int_0^{\pi} \frac{(\sin x - x \cos x)}{x^3} \cdot \cos \frac{x}{2} dx = \frac{3\pi}{16}$ 7